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Assignment

Networking with Windows Server

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25. Discuss the role of Windows Firewall in Windows Server and how to configure it.

🔐 Role of Windows Firewall in Windows Server

- Traffic Filtering: Monitors and controls network traffic entering or leaving the server.
- Protection Against Attacks: Blocks unauthorized access attempts, malware, and exploits.
- Segmentation: Allows administrators to define rules for different profiles (Domain, Private, Public).
- Compliance & Auditing: Helps meet security standards by logging and auditing traffic.
- Integration with Group Policy: Enables centralized management across multiple servers in a domain.

⚙️ Steps to Configure Windows Firewall

1. Access Firewall Settings

- Open Windows Firewall with Advanced Security from the Start menu or Control Panel.
- Alternatively, use Server Manager → Tools → Windows Firewall with Advanced Security.

2. Understand Profiles

- Domain Profile: Used when connected to a domain network.
- Private Profile: For trusted private networks.
- Public Profile: For untrusted networks (most restrictive).

3. Set Baseline Policies

- Define default inbound/outbound rules:
- Inbound: Block by default, allow only necessary services.
- Outbound: Allow by default, restrict if needed.

4. Create Firewall Rules

- Inbound Rules: Allow specific ports/services (e.g., HTTP on port 80, RDP on port 3389).
- Outbound Rules: Restrict traffic to trusted destinations.

Rules can be based on: Program, Port, Predefined services and Custom criteria (IP ranges, protocols)

5. Configure Logging & Auditing

- Enable logging to track dropped packets and successful connections.
- Store logs securely for compliance and troubleshooting.

6. Centralized Management (Optional)

- Use Group Policy Objects (GPOs) in Active Directory to apply firewall rules across multiple servers.
- This ensures consistency and simplifies administration.

7. Test & Validate

- Verify connectivity for allowed services.
- Use tools like ping or telnet to confirm rules work as intended.
- Periodically review and update rules to match evolving security needs.

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26. What is Network Address Translation (NAT) in Windows Server, and how do you configure it?

NAT translates private IP addresses (used inside a local network) into a public IP address for communication with external networks.

- Use Cases: mall offices with limited public Ips, Virtualized environments (Hyper-V) where multiple VMs share one external connection.

⚙️ Configuring NAT in Windows Server (RRAS Method)

1. Install RRAS Role
    - Open Server Manager → Add Roles and Features.
    - Select Remote Access → Routing role services.
2. Enable RRAS
    - Go to Server Manager → Tools → Routing and Remote Access.
    - Right-click the server name → Configure and Enable Routing and Remote Access.
3. Choose NAT Configuration
    - Select Network Address Translation (NAT) during setup.
    - Ensure the server has two interfaces:
    - One connected to the internal network (LAN).
    - One connected to the external network (Internet).
4. Configure NAT on External Interface
    - In RRAS console, expand IPv4 → NAT.

    - Right-click the external interface → Enable NAT.
    - This interface will translate private IPs to the public IP.

5. Verify Internal Connectivity

- Ensure DHCP is assigning private IPs to internal clients.
- Test Internet access from a client machine.

❁ Configuring NAT in Hyper-V (Virtualized Environment)

1. Enable Hyper-V
    - Ensure Hyper-V is installed and enabled on Windows Server.
2. Create NAT Virtual Switch
    - Use PowerShell:

New-VMSwitch -SwitchName "NATSwitch" -SwitchType Internal New-NetIPAddress -IPAddress 192.168.100.1 -PrefixLength 24 -InterfaceAlias "vEthernet (NATSwitch)" New-NetNat -Name "NATNetwork" -InternalIPInterfaceAddressPrefix 192.168.100.0/24

This creates a NAT network with the 192.168.100.0/24 subnet.

3. Attach VMs
    - Connect virtual machines to the NATSwitch.
    - Assign IPs from the NAT subnet (via DHCP or manually).
4. Test Connectivity

- VMs should now access the Internet through the host's public IP.

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27. Explain the concept of Dynamic Host Configuration Protocol (DHCP) and how to configure it in Windows Server 2016.

Dynamic Host Configuration Protocol (DHCP) in Windows Server 2016 automatically assigns IP addresses and other network configuration details to client devices, simplifying network management and reducing errors. Configuring DHCP involves installing the DHCP role, authorizing the server, and creating scopes to define IP ranges and options.

- **Definition:** DHCP is a network management protocol that automates the assignment of IP addresses, subnet masks, default gateways, and DNS servers to client devices.
- **Purpose:**
    - Eliminates manual IP configuration, Prevents IP conflicts, Efficiently manages IP address pools.

❁ **Steps to Configure DHCP in Windows Server 2016**

1. Install the DHCP Role

- Open Server Manager → Add Roles and Features.
- Select DHCP Server under server roles.
- Complete the installation wizard.

2. Authorize the DHCP Server

- After installation, open DHCP Manager.
- Right-click the server → Authorize.
- This step is required in Active Directory environments to prevent rogue DHCP servers.

3. Create a DHCP Scope

- A scope defines the range of IP addresses the DHCP server can assign.
- Steps:
- In DHCP Manager, right-click IPv4 → New Scope.
- Enter a name and description.
- Define the IP range (e.g., 192.168.1.100 – 192.168.1.200).
- Set subnet mask (e.g., 255.255.255.0).
- Configure exclusions if certain IPs should not be assigned.
- Set lease duration (default: 8 days).

4. Configure DHCP Options

- Add essential options for clients:
- Router (Default Gateway): e.g., 192.168.1.1
- DNS Servers: e.g., 8.8.8.8, 8.8.4.4
- Domain Name: e.g., office.local

5. Activate the Scope

- Once configured, activate the scope so clients can start receiving IP addresses.

6. Verify Functionality

- On a client device, release and renew IP (ipconfig /release and ipconfig /renew).
- Confirm the client receives an IP from the defined scope.

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28. Describe the configuring DNS (Domain Name System) in Windows Server.

DNS (Domain Name System) in Windows Server is used to translate human-friendly domain names (like example.com) into IP addresses that computers use to communicate. Configuring DNS ensures proper name resolution within a network and for external resources.

⚙️ Steps to Configure DNS in Windows Server

1. Install DNS Role
    - Open Server Manager → Add Roles and Features.
    - Select DNS Server and complete installation.
2. Open DNS Manager
    - Go to Server Manager → Tools → DNS.
    - This console is used to manage zones and records.
3. Create a Forward Lookup Zone
    - Right-click Forward Lookup Zones → New Zone.
    - Choose Primary Zone and define the domain name (e.g., office.local).
    - This zone maps domain names to IP addresses.
4. Add DNS Records
    - A Record (Host): Maps a hostname to an IP address.
    - CNAME Record: Creates an alias for another domain name.
    - MX Record: Defines mail server for the domain.
5. Configure Reverse Lookup Zone (Optional)
    - Maps IP addresses back to hostnames for troubleshooting.
6. Test Resolution

- Use nslookup or ping to verify that names resolve correctly.

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29. What is Server Manager, and how do you use it to manage servers in Windows Server?

Server Manager in Windows Server is a centralized management console that allows administrators to configure, monitor, and manage local and remote servers from a single interface. It simplifies server administration by providing a dashboard for roles, features, and performance.

🌐 Role of Server Manager

- Centralized Management: Manage multiple servers from one console.
- Role & Feature Deployment: Add or remove roles (e.g., DNS, DHCP, IIS) and features easily.
- Monitoring: View server health, performance, and events.
- Remote Administration: Connect to and manage other servers in the network.
- Configuration: Provides wizards for setting up services and roles quickly.

⚙ How to Use Server Manager

1. Open Server Manager
    - It launches automatically after login, or you can open it from the Start menu.
2. Add Servers
    - Use Manage → Add Servers to connect remote servers by name, IP, or Active Directory.
3. Install Roles and Features
    - Navigate to Manage → Add Roles and Features.
    - Follow the wizard to install services like DNS, DHCP, or File Services.
4. Monitor Server Performance
    - Use the dashboard to check alerts, events, and performance metrics.
    - Quickly identify issues across multiple servers.
5. Configure Services
    - Access role-specific management tools directly from Server Manager.
    - For example, configure DHCP scopes or DNS zones after installing those roles.
6. Group Servers

- Organize servers into groups for easier monitoring and management.

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30. Discuss the role of Remote Desktop Services (RDS) in Windows Server 2016 or 2019 and how to configure it.

Remote Desktop Services (RDS) in Windows Server 2016/2019 provides centralized access to applications and desktops, allowing users to connect remotely to a server or virtual desktop environment. It's widely used in organizations to enable secure remote work, application virtualization, and centralized management of resources.

⚙ Steps to Configure RDS in Windows Server 2016/2019

1. Install RDS Role

- Open Server Manager → Manage → Add Roles and Features.
- Select Remote Desktop Services installation.
- Choose Standard Deployment (for multiple servers) or Quick Start (single server).

2. Deploy RDS Components

- RD Session Host: Hosts applications and desktops.
- RD Connection Broker: Manages load balancing and reconnections.

- RD Web Access: Provides access via a web browser.
- RD Licensing: Manages RDS CALs (Client Access Licenses).

3. Configure Licensing

- In Server Manager → Remote Desktop Services → Overview, add the RD Licensing server.
- Activate the license server and install RDS CALs.

4. Publish Applications or Desktops

- Use Server Manager → Remote Desktop Services → Collections.
- Create a new collection to group session hosts.
- Publish applications (RemoteApp) or full desktops for users.

5. Configure User Access

- Assign users or groups to the RDS collection.
- Ensure proper permissions are set in Active Directory.

6. Test Connectivity

- Users can connect via Remote Desktop Client (mstsc) or RD Web Access portal.
- Verify applications and desktops launch correctly.